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# er's 12 ciples



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Richard Mayer's multimedia learning theory is a must-read for instructional designers, eLearning developers and L&D professionals everywhere. Mayer's principles of multimedia learning provide a blueprint for how to structure multimedia elements to maximise learning outcomes.

A distinguished professor of psychology at the University of California, Mayer published his cognitive theory of multimedia learning in 2001. And the principles he developed after several years of research are just as relevant today. From images and video to AR and VR, multimedia is now integral to digital education. And learners find it a more engaging and enjoyable way to learn. According to one survey, 70% of students prefer digital learning to traditional classrooms.

Mayer's multimedia learning theory is based on three assumptions:

- **Dual-channel assumption:** According to Mayer, people have two separate channels for processing auditory and visual information.
- **Limited-capacity assumption:** The theory recognises that individuals have a limited ability to absorb information at any one time.
- **Active-processing assumption:** The multimedia learning theory suggests that people should be actively engaged in the learning process rather than passive

receivers of information.

From these assumptions, Mayer goes on to identify 12 principles of multimedia learning. And these principles provide an invaluable checklist for designers wanting to optimise learning with multimedia.

Today's post examines Mayer's principles of multimedia learning and shares some practical ways they can be incorporated into eLearning.


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| Mayer's 12 Principles at a Glance | |
|---|---|
| Multimedia | Use words + pictures |
| Coherence | Remove unnecessary content |
| Signalling | Highlight what matters |

| | |
|---|---|
| Redundancy | Avoid duplicate text + narration |
| Spatial Contiguity | Keep related items close together |
| Temporal Contiguity | Present words + visuals together |
| Segmenting | Break content into chunks |
| Pre-training | Teach basics first |
| Modality | Use narration with visuals |
| Voice | Use human voiceovers |
| Personalization | Use conversational tone |
| Image | Talking heads are not always needed |

# What are Mayer's 12 principles of multimedia learning?

The principles are grounded in cognitive science and how people process information. They provide a checklist on how to structure multimedia learning experiences.

Tick off the following principles as you design your program to ensure you maximise learner comprehension, improve retention and enhance the learning outcomes.

## 1. Multimedia Principle

**What it means:** People learn best from a combination of words and pictures. Instructional designers should use words (text or narration) and visuals (images, animations, or videos) rather than only one channel. Presenting information in multiple formats helps learners process and integrate information more effectively.

**How to apply the multimedia principle:**

- Use a mix of text and images.
- Incorporate visuals to illustrate key points in the eLearning program.
- Instead of using images for the sake of it, double-check that the visuals clarify meaning or enhance comprehension.

## 2. Coherence Principle

**What it means:** Learning is more effective if unnecessary information is excluded rather than included. eLearning developers should ensure that words and visuals are closely aligned and complement each other. Do away with irrelevant information or fluff that might distract learners from the main message.

**How to apply the coherence principle:**

- Only include graphics, text or narratives if they are on point and support the learning goals.
- Avoid using background music.
- Use simple diagrams and infographics.

## 3. Signalling Principle

**What it means:** Learning is enhanced when cues are added to draw attention to vital information. Online learning designers should make it easy for students by highlighting what's important. Too much information on the screen confuses the learner, making it harder to work out the most critical elements.

**How to apply the signalling principle:**

- Emphasise key points with arrows, callouts, highlights or bold text.

## 4. Redundancy Principle

**What it means:** The redundancy principle suggests that we learn best from a combination of spoken words and graphics. Add on-screen text, and you risk overwhelming students. Therefore, designers should avoid presenting the same information in multiple formats simultaneously. Redundant information can create overload and gets in the way of learning.

**How to apply the redundancy principle:**

- Use either graphics or text to complement spoken presentations. Never use both at the same time.
- Minimise the use of on-screen text in narrated presentations. Instead, focus on images or graphics.

## 5. Spatial Contiguity Principle

**What it means:** Mayer says text and visuals should be presented close together on the screen to maximise learning. L&D professionals should align visuals and text, so learners can more easily understand the relationships between them. Avoid spatially separating text from related graphics or animations.

**How to apply the spatial contiguity principle:**

- Keep text and visuals close to each other in the frame.
- Place any feedback next to the relevant questions or answers.
- Ensure directions are presented on the same screen as an activity.

## 6. Temporal Contiguity Principle

**What it means:** This principle suggests that students learn best when words and pictures are presented at the same time rather than sequentially. Simultaneous presentation allows learners to process the information together and build meaningful connections. For example, students shouldn't learn about a process and then watch an animation about it afterwards. Instead, designers should ensure the voiceover plays along with the animation.

**How to apply the temporal contiguity principle:**

- Ensure voiceovers are timed with visuals or animations.
- Place related text and pictures on the same screen.

## 7. Segmenting Principle

**What it means:** Mayer found that better learning outcomes are achieved when

information is segmented, and students have control over the pace. For developers, this means breaking down complex information into smaller, manageable chunks. Present the information in a step-by-step approach, allowing learners to process each segment independently and build understanding gradually.

**How to apply the segmenting principle:**

- Organise content in manageable, coherent bite-sized chunks.
- Ensure no one lesson, module, or slide has too much information packed in.
- Allow users to control the pace of instruction with next buttons or speed controls.

## 8. Pre-training Principle

**What it means:** When it comes to multimedia learning, this principle states that people learn better when they already know the basics. Often, this means understanding definitions, terms or critical concepts before diving into the details. For example, you can't expect a student to complete a task using Excel if they have no experience in the software.

Instructional designers should give learners an overview of key concepts before presenting the main content. Pre-training activates prior knowledge and primes learners to understand better and retain new information.

**How to apply the pre-training principle:**

- Develop an introductory module to explain key concepts before starting the main program.
- Consider preparing a cheat sheet of terms and definitions to accompany the course.
- Ensure students know how to use any tools needed to complete tasks within the course.

## 9. Modality Principle

**What it means:** The modality principle says that students experience deeper learning from visuals and spoken words than text and visuals. This doesn't mean you shouldn't have text on the screen. It's more about ensuring a balance, as too much text can overwhelm students.

Designers should use visual and auditory channels based on the content and the learner's preferences. Consider using animations or images to illustrate dynamic processes and narration to explain complex concepts.

**How to apply the modality principle:**

- Try to limit your use of text. Instead, rely on visuals, images and voice overs.
- During a narrated presentation with visuals, only use text to list steps or provide directions.

## 10. Voice Principle

**What it means:** This principle is straightforward. People learn better when real presenters rather than machines make voice overs. Although we are all used to Siri and Alexa, it seems we still prefer a friendly, human touch.

**How to apply the voice principle:**

- This one is simple. Narrate your own audio content or use a voiceover professional.
- If doing it yourself, ensure you have a high-quality microphone and use audio editing software.

## 11. Personalisation Principle

**What it means:** The personalisation principle is another common sense one. Learning with multimedia works best when it's personalised and focused on the user. For designers, this means speaking in the first person (I, you, we, our). Avoid formal language and instead use a conversational tone to engage learners. Imagine you are

in the room speaking with students.

**How to apply the personalisation principle**

- Use accessible, everyday language in your content.
- Consider the demographics of your target audience and tailor your language accordingly.

## 12. Image Principle

**What it means:** Mayer points out that the research is still in its early stages. However, the image principle suggests people may not learn better from talking head videos. High-quality, complementary visuals can often be more effective than having a speaker's image.

- Consider using talking head videos initially to develop connections and build trust only.
- After that, select relevant and meaningful images that align with the instructional content.

# Multimedia Learning Theory: final thoughts

Mayer's 12 principles of multimedia learning are an essential resource for instructional designers and eLearning developers. Use them to guide your course development and get the most out of multimedia materials. That way, you will enhance learner engagement, comprehension, and retention, leading to improved outcomes for your organization.

# FAQs: Mayer's Multimedia Learning